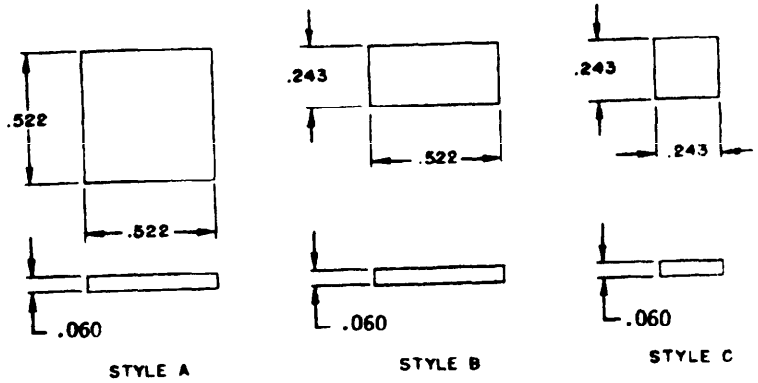
MILITARY SPECIFICATION SHEET

SWITCH, PUSH BUTTON, ILLUMINATED

COLOR FILTER

The complete requirements for procuring the color filter described herein shall consist of this document and the latest issue of Specification MIL-S-22885.



	INCHES	MM
	0.055	1.40
(B)	.060	1.52
	.243	6.17
	.522	13.26

NOTES:
1. Dimensions are in inches.
2. Tolerance is ±.005(.13 mm).
3. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

(B) Denotes changes.

FSC 5930

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REQUIREMENTS:

Material:	Acrylic per MIL-P-5425.
Design and construction:	See figure.
Operating temperature:	-55°C to +60°C.
Intended use:	For use with switch in accordance with MIL-S-22885/67.
Part numbers:	See Table II.

Inspection:

Group A	-Inspection for dimensions.
Group B	-Color comparison.

Measurement of color and photometric brightness:

Photometric brightness: The measuring apparatus shall consist of a suitable brightness meter or other brightness measuring device which will accurately measure the brightness of the illuminated diffusing lens. The front of the surface of the illuminated module shall be perpendicular to the line of sight of the brightness meter. The lamps (midget flange T-1 3/4 industry number 328 incandescent) shall be aged and tested and shall be operated at candlepower (0.34 ± 0.02 scp). This shall be verified before and after the tests. Tests shall be conducted in a dark room. A minimum of five readings shall be taken to enable a valid record to be made for qualification inspection. One reading at center shall be taken for group B inspection. If a visual brightness meter is used, the comparison field for the measuring instrument shall be a near color match to the test area being measured. If a physical (photoelectric) brightness photometer is used, the light receptor apparatus must have a spectral sensitivity which matches the luminous efficiency curve of the CIE standard observer, or must be calibrated to measure the luminance (photometric brightness) or the colored light.

Colors: All colors shall be referenced within certain limits established upon the CIE chromaticity diagram recommended by the International Commision on Illumination in 1933. Acceptable illuminated colors shall lie within the limiting areas defined by the points shown in table I.

Chromaticity test: For qualification, the spectographic method shall be used and three copies of the actual plotted measurement obtained for each color shall be forwarded to the activity responsible for qualification.

For group B inspection, the chromaticity of light emitted by the sample equipment may be determined by comparison with light of known chromaticity. A set of filters of known luminous transmittance and of spectral characteristics may be used in conjunction with a visual comparison to test for conformity to the chromaticity requirements. Comparison methods may be applied directly to either the assembled equipment, filters, or a flat filter sample of representative wall thickness and lot color. In cases of dispute, actual x and y coordinates shall be determined by spectrographic methods.

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Table I - Chromaticity coordinates.

Chromaticity coordinates 1/ 2200° Kelvin		
Colors	X	Y
Green	0.312	0.685
	.312	.630
	.185	.630
	.265	.720
Yellow	.530	.442 2/
	.544	SL 2/
	.571	.402 2/
	.584	SL 2/
Red	.680	.300
	.690	.310
	.715	.265
	.725	.275
Blue	.200	.260
	.120	.220
	.085	.280
	.175	.315
Amber	.565	SL 2/
	.565	.405
	.605	SL 2/
	.605	.365
White	.484 Ref.	.415 Ref.
X is not less than .350. Y is not greater than .470. $Y - Y_o$ or $Y_o - Y$ is not numerically greater than 0.03X. (Y_o is the Y coordinate of the Planckian radiator for a given X).		

(B)

1/ The chromaticities of the diffusing lenses or colored filters expressed as X and Y coordinates on the CIE chromaticity diagram shall lie within areas bounded by the coordinates listed for each color, using a source illuminate of 2200°K.
2/ SL = Spectrum Locus (where intersected by other of coordinate pair).

Table II - Part numbers

Part Number	Color	Code	Style
M22885/70-001	Blue	B	A
M22885/70-002	Green	G	A
M22885/70-003	Red	R	A
M22885/70-004	White	W	A
M22885/70-005	Yellow	Y	A
M22885/70-006	Amber	A	A
M22885/70-007	Blue	B	B
M22885/70-008	Green	G	B
M22885/70-009	Red	R	B
M22885/70-010	White	W	B
M22885/70-011	Yellow	Y	B
M22885/70-012	Amber	A	B
M22885/70-013	Blue	B	C
M22885/70-014	Green	G	C
M22885/70-015	Red	R	C
M22885/70-016	White	W	C
M22885/70-017	Yellow	Y	C
M22885/70-018	Amber	A	C

Preparing activity:
Navy - EC
(Project 5930-N297-2)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

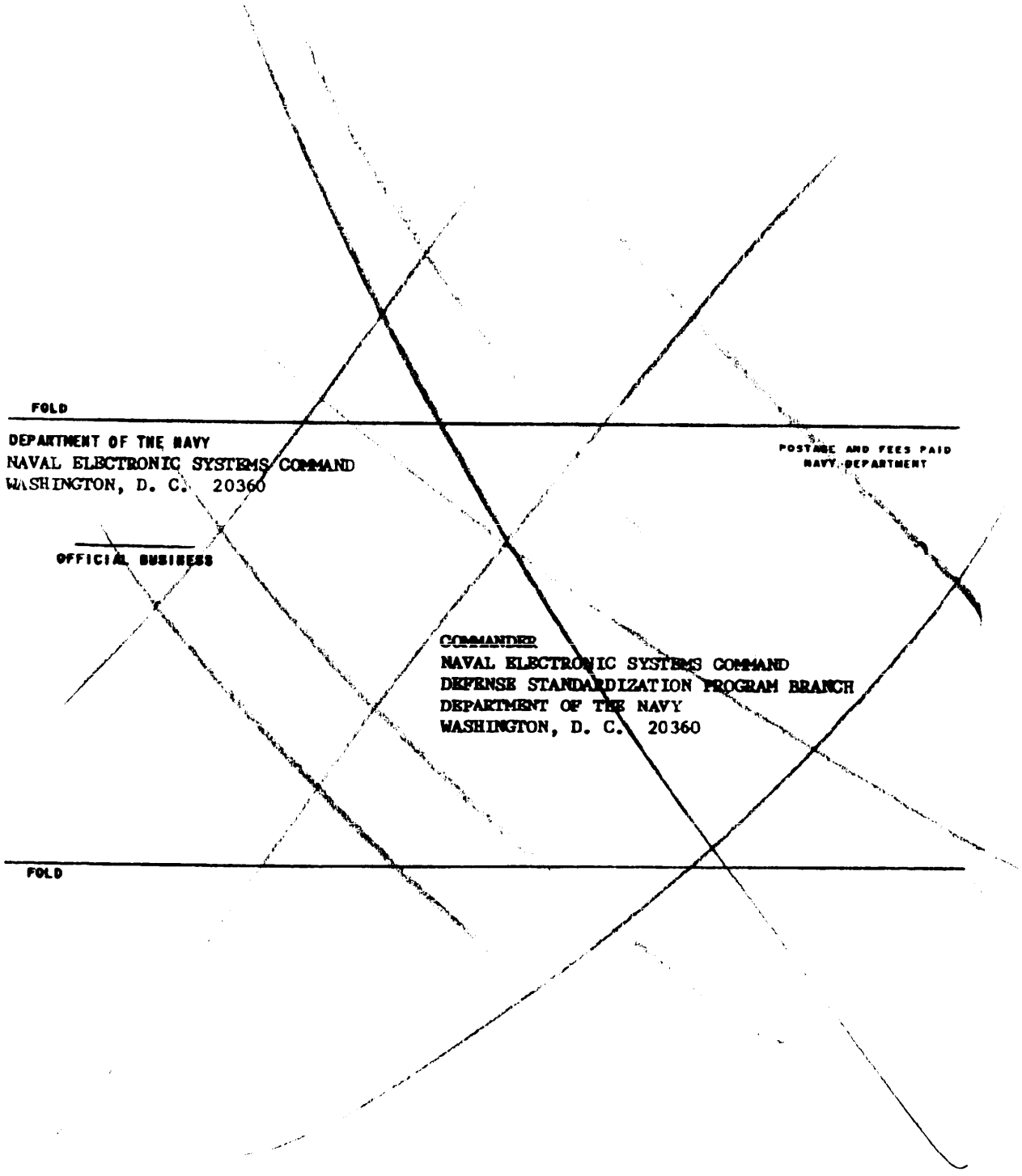
☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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